

Phase Morphology and Conductive Properties of PBT/POE-*g*-GMA/PP/CNT Nanocomposites with a Tri-Continuous Structure via Thermal Annealing

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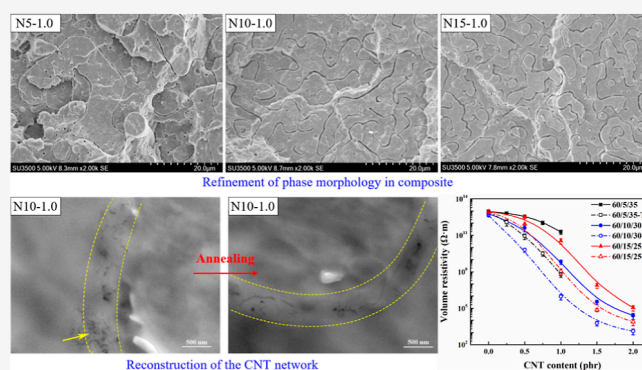
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ABSTRACT: In order to improve the conductivity of polymer composites and elucidate the relationship between the microstructure and conductivity, a series of PBT/POE-*g*-GMA/PP/CNT nanocomposites with a tri-continuous structure and low conductive percolation threshold were prepared by melt blending. Compared to PBT/PP/CNT composites, the PBT/POE-*g*-GMA/PP/CNT nanocomposites show better conductivity owing to the selective distribution of carbon nanotubes (CNTs) in the POE-*g*-GMA phase with the same CNT content. The increasing content of the CNTs and POE-*g*-GMA refines phase morphology, thus influencing the CNTs' conductive path. Composites containing 10 wt % POE-*g*-GMA display the lowest conductive percolation threshold at 1.0 phr. In addition, the influencing mechanism of thermal annealing on microstructure and conductivity was methodically investigated. Thermal annealing can effectively enhance the rebuilding of CNT conductive networks and continuity of the POE-*g*-GMA phase on the premise of the original phase structure, decreasing the volume resistivity of nanocomposites. The nanocomposites obtained in this study possess excellent electrical properties alongside the potential to be a candidate material in fields of sensors, electromagnetic interference shielding, and so forth.



1. INTRODUCTION

Over the past decades, conductive polymer composites have been widely used for their advantages of low cost, good processability, and tunable properties. The double percolation theory is one of the major theories in this domain. Conductive fillers are selectively distributed in the continuous phase or at the phase interface.^{1–4} Due to their distribution selectivity, the actual local effective concentration of fillers increases in composites, resulting in good electrical conductivity with low filler content.⁵ However, it remains a significant challenge to perform a precise distribution of fillers in co-continuous composites.

Currently, researchers have carried out the interface distribution of conductive fillers by two means: kinetic regulation and thermodynamic regulation. Kinetic regulation dominates the migration of fillers by changing the sequence or time of melting, mixing fillers and the polymers with low compatibility first, then with good affinity, to make the conductive fillers migrate to a polymer of good affinity in melt blending due to the difference in affinity.^{6–9} Thermodynamic regulation grafts block copolymers or random copolymers onto conductive fillers to make the modified fillers interact with both polymers in composites.^{10–12} The degree of modification has an impact on the balance of two interactions and the

distribution of fillers. However, both methods present some challenges in the actual process. Dynamic regulation makes it difficult to perform fillers' precise migration to the interface, especially for high-aspect ratios fillers such as carbon nanotubes (CNTs) due to their low interface stability and high migration rate. In our earlier study, the migration of carboxyl-modified CNTs was observed in the preparation of PET/POE-*g*-GMA/CNT nanocomposites due to the interaction between carboxylic and epoxy groups in POE-*g*-GMA, which occurred in the part of the CNTs distributed in the POE-*g*-GMA phase. In addition, the structure of composites by dynamics is usually in a state of thermodynamic non-equilibrium, which can affect the distribution of fillers and then destroy the filler networks in the subsequent process.¹³ In terms of thermodynamics, it is complicated to prepare amphipathic fillers in large-scale industrial production, and

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grafting may destroy the original filler structure, further affecting conductivity.

Recently, a new method has been proposed to control the interface distribution of fillers of the conductive composites by introducing a third polymer component and constructing a tri-continuous structure in composites.^{14–17} Among the tri-continuous blends, two principal component polymers form a co-continuous structure; the third component polymer is distributed at the phase interface of the two main polymers, exhibiting a continuous phase, and the fillers are selectively within the third component polymers. This method simply restricts fillers in the continuous phase of the middle, thus obtaining the percolation in the composites and solving the problem that the fillers are difficult to distribute precisely at the polymer interface. Cohen et al.¹⁸ prepared a tri-continuous blend of polyamide (PA), polypropylene (PP), and ethylene-methyl acrylic acid (EMAA) via melt blending, in which EMAA as the middle continuous phase was dispersed at the PA/PP-phase interface. CNTs are selectively distributed in the EMAA phase and limited at the PA/PP phase interface because of π – π interaction between EMAA and CNTs. In comparison with PA/PP/CNT co-continuous structural composites, the volume resistivity of the tri-continuous structural composites decreases by approximately 4 orders of magnitude with the same CNT content (1% by weight).

Studies have shown that annealing could improve the conductivity of conductive composites.^{19–21} Cipriano et al.²² prepared polystyrene (PS)/CNTs and PS/carbon fiber composites and found that the conductivity of composites was significantly improved after annealing. For co-continuous composites, Pan et al.²³ improved the conductivity of PS/polymethyl methacrylate/CB composites by annealing, which indicated that annealing can significantly enhance the conductivity of composites. However, for the tri-continuous structural composites, the function and impact of annealing are not clear. Therefore, it is worth exploring whether annealing can further improve the conductivity of composites and how it will affect the interface distribution of CNTs.

In this study, we prepared the PBT/POE-g-GMA/PP/CNT composites with a tri-continuous structure using a melt blending method, in which PBT and PP were the main polymer phases, POE-g-GMA was the third component, and carboxyl-modified CNTs were the conductive fillers distributed at the PBT/PP interphase. The objective of this study is to comprehensively explore the effect of the conductive filler and the third component content on the phase morphology and electrical conductivity of composites and then to investigate the mechanics of thermal annealing on the morphology and conductivity of composites in depth.

2. EXPERIMENTAL SECTION

2.1. Materials. PBT (S600F10) was purchased from Dupont (USA). POE-g-GMA containing 1.5 wt % GMA was obtained from Fine-Blend Compatilizer Co., Ltd. (China). PP was provided by LyondellBasell (The Netherlands). Modified CNTs (TNIMC4) containing 1.55 wt % carboxyl groups were supplied by the Chengdu Institute of Organic Chemistry, Chinese Academy of Sciences (China). The outer diameter of CNTs was 10–30 nm, and the length was 10–30 μ m. Xylene was obtained from Sinopharm Chemical Reagent Co., Ltd. (China). *m*-Cresol was provided by Shanghai McLean Biochemical Technology Co., Ltd.

2.2. Sample Preparation. Prior to melt blending, all materials were dried in a vacuum oven. PBT was dried at 120 °C for 4 h, and PP and POE-g-GMA were dried at 60 °C for 6 h. In order to maximize the distribution of CNTs in the third component POE-g-GMA, the POE-g-GMA/CNT master batches were prepared by mixing the dried POE-g-GMA and 20 phr CNTs homogeneously in proportion and then melt-blending in a single-screw extruder (Brabender Measurement and Control System, Brabender Instrument Company, Germany). The extruder temperature was 180–200 °C, and the screw speed was 50 rpm. The vacuum-dried extruded pellets were blended with dried PBT, PP, and POE-g-GMA in proportion and then melt-blended to prepare PBT/POE-g-GMA/PP/CNT nanocomposites by a twin-screw extruder (HAAKE PolyLab OS, Thermo Electron Karlsruhe GmbH, Germany). The temperature of the extruder was 230–250 °C, and the screw speed was 300 rpm. The mass ratios of PBT/POE-g-GMA/PP in blends were 60/5/35, 60/10/30, and 60/15/25. The content of CNTs in the nanocomposites was 0–1 phr for the restriction by masterbatch preparation, and the maximum addition amount was 1 phr. For the PBT/POE-g-GMA/PP (60/10/30) and PBT/POE-g-GMA/PP (60/15/25), the content of CNTs was 0–2 phr. The extruded pellets after vacuum drying were processed into test specimens by a hot-press machine (GT-7014-A50C, Gotech, China). The hot-pressing temperature was 240 °C. The thermal annealing treatment was carried out by hot-pressing in the mold corresponding to 240 °C and 10 MPa for 1 h.

For convenience, the PBT/POE-g-GMA/PP/CNT nanocomposites with different contents were named N_x-y , respectively, where x represents the POE-g-GMA content and y represents the CNT content in the sample. For example, the N5-0 sample meant the PBT/POE-g-GMA/PP (60/5/35) blend without CNTs, and the N5-0.5 sample meant the PBT/POE-g-GMA/PP (60/5/35) nanocomposites with 0.5 phr CNTs. The annealed sample was labeled N_x-y-T .

2.3. Morphology Characterization. The phase morphology of the samples was characterized by a field emission scanning electron microscope (SU-3500 and SU-8010, Hitachi, Japan) after soaking in liquid nitrogen and quenching. Some samples were selectively etched by xylene at 75 °C for 2 h to dissolve POE-g-GMA and some at 140 °C for 2 h to remove PP and POE-g-GMA. The etching of PBT was performed using *m*-cresol at 130 °C for 2 h. Samples were vacuum-dried and gold-coated for SEM testing. The average thickness of PBT or POE-g-GMA was calculated by image analysis software (Image-Pro PLUS). The distribution of CNTs in the samples was characterized by transmission electron microscopy (TEM) (HT-7700, Hitachi, Japan). The microstructure of the specimens was observed using an optical microscope (ECLIPSE E600W POL, NIKON, Japan).

2.4. Continuity Test. The continuity of the PBT and POE-g-GMA phase in the nanocomposites was tested by selective solvent extraction experiments. A sample of around 0.2 g was dried in a vacuum oven and then weighed as m_1 . The dried sample was immersed in a large quantity of *m*-cresol for PBT or in xylene for POE-g-GMA and then magnetically stirred to completely dissolve the PBT or POE-g-GMA phase. The extracted specimen was dried in a vacuum oven and weighed as m_2 . The continuity of the POE-g-GMA phase was calculated in accordance with eq 1

$$\text{continuity}(\text{POE} - g - \text{GMA}) = \frac{m_1 - m_2}{m_1(\text{POE} - g - \text{GMA})} \times 100\% \quad (1)$$

where $m_1(\text{POE-g-GMA})$ is the mass of the POE-g-GMA phase in the sample before extraction.

2.5. Electrical Resistivity Test. The electrical resistivity was obtained on a Keithley 6517A electrometer (Keithley Instruments, Inc.) connected with a Keithley 8009 resistivity test fixture. The diameter of the tested samples was 90 mm, and the thickness was 2 mm. The average of five tests for each sample was taken as the result.

3. RESULTS AND DISCUSSION

3.1. Selective Location of the CNTs in Nanocomposites. The selective distribution of CNTs in nanocomposites can be observed by scanning electron microscopy (SEM), TEM, and optical microscopy (OM), as illustrated in Figure 1. In Figure 1a, POE-g-GMA in sample N10-1.0 was

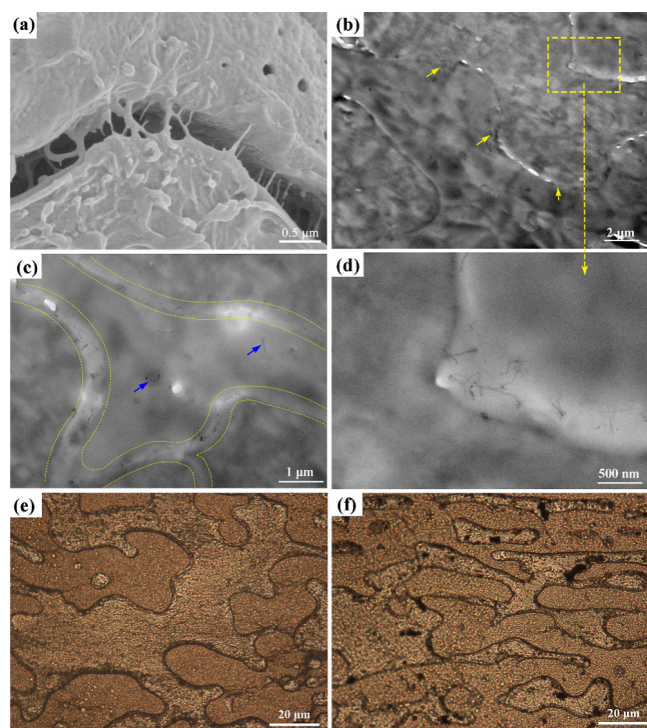


Figure 1. SEM, TEM, and OM pictures of the N10 nanocomposites: (a) N10-1.0; (b,c,d) N10-0.5; (e) N10-0; (f) N10-1.0.

etched. As noted, CNTs are distributed at the PBT/PP phase interface, where POE-g-GMA is located in the original composites. Since some CNTs are partially inserted into PBT or PP phases, they cannot be dissolved in a xylene solution with POE-g-GMA. For Figure 1b, PBT and PP exhibit a co-continuous structure and CNTs are distributed within the PBT/PP interface area, as indicated by the yellow arrows. As shown in the high-magnification TEM diagram in Figure 1c,d, POE-g-GMA is continuously located along the PBT/PP interface, with most of the CNTs distributed within. Only a few CNTs are dispersed in PBT and PP phases (indicated by the blue arrows). The selective distribution of CNTs is further proven in Figure 1e,f, in the OM diagram. N10-0 composites without CNTs possess a tri-continuous structure, with a

continuous POE-g-GMA thin layer separated at the PBT/PP interface. Because of the lowest transmittance of aggregated CNTs in the POE-g-GMA layer, the color of the POE-g-GMA continuous layer at the PBT/PP phase interface is darker in N10-1.0. Based on the results above, most CNTs are located in the POE-g-GMA phase and their distribution selectivity in POE-g-GMA continuous thin layer facilitates the construction of continuous conductive paths at a low concentration, enhancing the electrical conductivity of nanocomposites.

3.2. Phase Morphology of the Tri-Continuous Nanocomposite. The phase morphology of tri-continuous nanocomposites with 1.0 phr CNTs was observed by SEM (Figure 2). Samples in Figure 2a–d were etched by xylene at 75 °C to

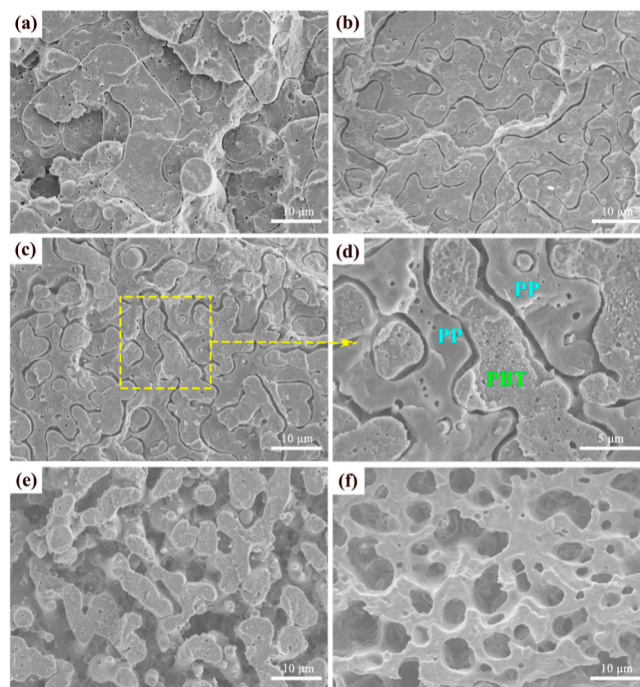


Figure 2. SEM images of the nanocomposites with 1.0 phr CNTs at different polymer ratios: (a) N5-1.0; (b) N10-1.0; (c) (d) N15-1.0; (e) N15-1.0, etched by xylene at 140 °C; (f) N15-1.0, etched by *m*-cresol.

selectively remove the POE-g-GMA phase. It can be observed that an obvious tri-continuous structure formed with POE-g-GMA located along the PBT and PP interface. As seen in the high-magnification SEM diagram of Figure 2d, most POE-g-GMA are scattered at the PBT/PP interface, forming a continuous thin layer with only a few localized in PBT and PP phases. To fully characterize tri-continuous composites, SEM was carried out after etching PP and POE-g-GMA with xylene at 140 °C or PBT with *m*-cresol. Figure 2e displays SEM images of N15-1.0 after etching POE-g-GMA and PP, and the remaining PBT is continuous. Figure 2f displays N15-1.0 after etching PBT, and PP and POE-g-GMA retained are continuous, too. Consequently, the tri-continuous layer structure is successfully established in the PBT/POE-g-GMA/CNT nanocomposites based on the analysis above.

Samples in Figure 2 all show typical tri-continuous structures, but there are certain differences in morphology. The phase morphology of composites gets refined and the thickness of the POE-g-GMA layer increases gradually as the POE-g-GMA content increases. The thickness of the POE-g-

GMA layer was calculated by Image-Pro PLUS image analysis, and results are presented in Table 1. The POE-g-GMA layer in

Table 1. Thickness of the POE-g-GMA Layer and Continuity of POE-g-GMA and PBT

sample	thickness of POE-g-GMA layer (μm)	continuity of POE-g-GMA (%)	continuity of PBT (%)
N5-1.0	0.314	80.7	83.6
N10-1.0	0.437	84.6	88.4
N15-1.0	0.623	85.4	91.3

the N15-1.0 sample has the thickest POE-g-GMA layer measuring $0.623\ \mu\text{m}$. We then verify the continuity of the POE-g-GMA and PBT phases, as illustrated in Table 1. The continuity of the POE-g-GMA phase of all specimens is over 80%, which indicates that most POE-g-GMA are situated at the PBT/PP interface and form a continuous layer. Compared to N5-1.0, due to the high POE-g-GMA content covering the PBT/PP interface, N10-1.0 and N15-1.0 have an increased continuity and a more refined POE-g-GMA continuous layer. Furthermore, the continuity of the POE-g-GMA is affected by the continuity of PBT. For instance, the continuity of PBT in N5-1.0 is inferior to that of N10-1.0 and N5-1.0, leading to a lower continuity of POE-g-GMA within.

SEM diagrams of PBT/POE-g-GMA/PP (60/30/10) with different CNT content are shown in Figure 3; samples were all

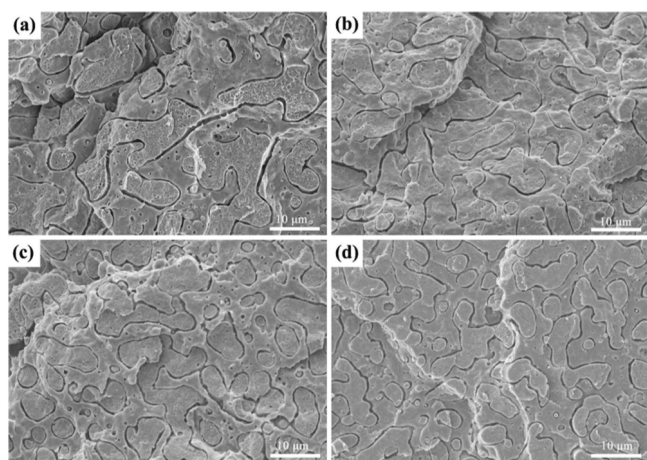


Figure 3. SEM images of the N10 nanocomposites with different CNT contents: (a) N10-0; (b) N10-0.5; (c) N10-1.5; (d) N10-2.0.

etched by xylene at $75\ ^\circ\text{C}$. A typical tri-continuous structure can be observed in N10-0 without CNTs (Figure 3a), and the continuous POE-g-GMA thin layer is located at the PBT/PP interface. With the addition of CNTs, the morphology of the composites gets refined. On the one hand, the CNTs located at the PBT/PP phase interface can physically block the polymer aggregation during melt blending. On the other hand, the viscosity of the POE-g-GMA phase increases as the CNT content increases, which can enhance the shear strength in the polymer during melt blending.

3.3. Electrical Properties of the Tri-continuous Nanocomposites. The volume resistivity of PBT/POE-g-GMA/PP composites with different CNT contents is illustrated in Figure 4. In our previous study, CNTs are located in the continuous PBT phase of PBT/PP composites. The PBT/PP blends (7/3)

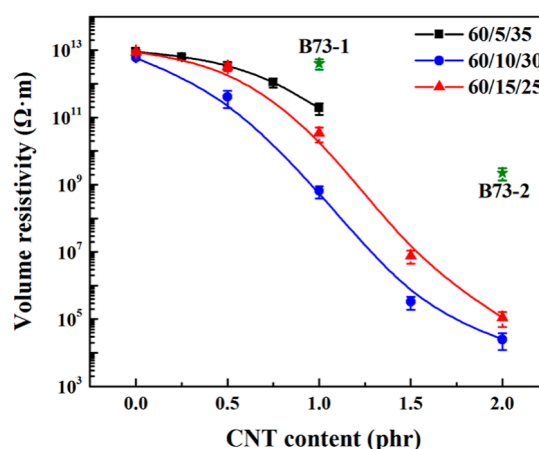


Figure 4. Volume resistivity of the nanocomposites as a function of the CNT content.

with different CNT contents are named as B73-*x*, as shown in Figure 4. B73-1 was still an insulator, and the volume resistivity of B73-2 was $2.2 \times 10^9\ \Omega\cdot\text{m}$. Compared to B73-1 and B73-2, the tri-continuous composites have a lower volume resistivity with the same CNT contents. In addition, CNTs are distributed at a continuous POE-g-GMA thin layer in the tri-continuous composites, effectively promoting the mutual contact between CNTs and establishing CNTs' conductive path in the POE-g-GMA layer. Therefore, the multilayered tri-continuous structure composites obtain excellent conductivity at a relatively low CNT addition.

The volume resistivity of composites with different POE-g-GMA contents is shown in Figure 4. With the increase of the POE-g-GMA content, the volume resistivity of composites first decreases and then increases because the CNTs' conductivity path is strongly affected by the continuity of the POE-g-GMA thin layer, related to the phase morphology of the composites. For the NS series, the continuity of the POE-g-GMA thin layer is lower than that of the N10 and N15 series due to lower POE-g-GMA content (Table 1). With the increase of POE-g-GMA content, the POE-g-GMA layer is constructed and refined, inducing the formed CNT networks along the POE-g-GMA layer, thus resulting in a low volume resistivity of the N10 series. However, the volume resistivity of the N15 series increases with the increase of POE-g-GMA content (Figure 4), which thickens the POE-g-GMA layer (Figure 2 and Table 1) and results in a decrease in CNTs' effective content. In addition, the phase morphology gets refined with the increase of POE-g-GMA, leading to the increased interface area and high percolation threshold. Therefore, the N15 series have a more refined phase morphology and thicker POE-g-GMA layer than that in the N10 series, resulting in low intensity of CNT networks and high volume resistivity.

3.4. Effect of Thermal Annealing on the Phase Morphology and Electrical Properties of Tri-Continuous-Structure Nanocomposites. According to previous research, high-aspect-ratio fillers like CNTs are difficult to distribute at the polymer interface with precision and stability during melt blending. In our previous study, annealing can lead to enhanced conductivity because of the reconstruction of CNT networks and the establishment of a better co-continuous structure. Therefore, the influence of thermal annealing on the phase morphology and conductivity of tri-continuous structural composites and the relationship between

the phase morphology and conductive properties are detailly explored.

The TEM images of N10-1.0 nanocomposites before and after annealing are shown in Figure 5. It is apparent that CNTs

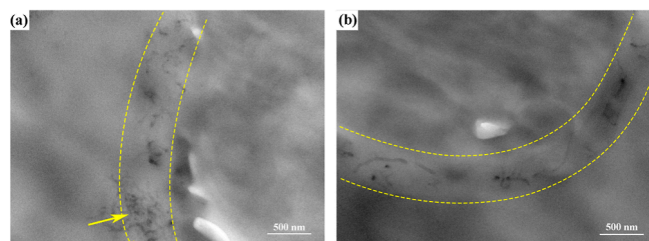


Figure 5. TEM images of the N10-1.0 nanocomposites before annealing (a) and after annealing (b).

are all located in the POE-g-GMA layer, at the PBT/PP interface, indicating their stability during annealing. Moreover, CNTs exhibit a homogeneous distribution after thermal annealing, which has been reported in similar studies.^{19–21,24} In the process of thermal annealing, polymer molecular chains are released to disperse CNT agglomerates, and CNTs are rearranged to construct homogeneous CNT networks.

Figure 6 shows the SEM images of composites with 10 phr CNTs after annealing, which were dissolved in xylene at 75 °C to eliminate POE-g-GMA. As indicated in Figure 6, the tri-continuous structure remains after annealing. Serious phase

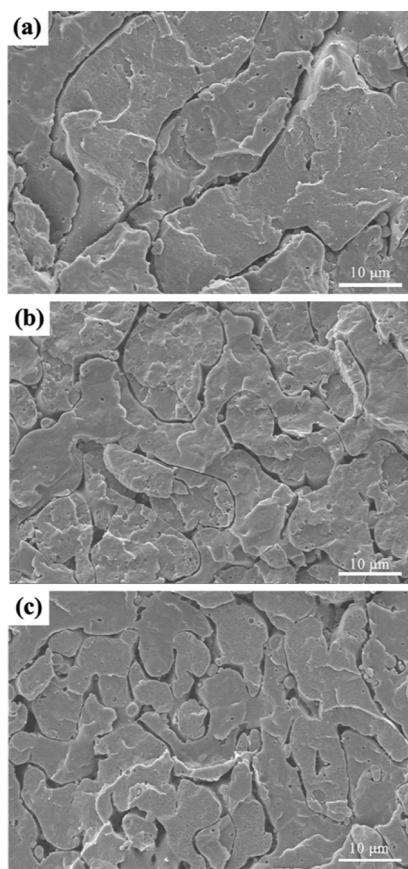


Figure 6. SEM images of the annealed nanocomposites with 1.0 phr CNTs at different polymer ratios: (a) N5-1.0; (b) N10-1.0; (c) N15-1.0.

coarsening and thickening of the POE-g-GMA layer occurs by annealing because the polymer aggregated to reduce interface area and reach system equilibrium. The thickness of the POE-g-GMA layer in samples after annealing is calculated and compared with those before annealing (Figure 7). For N5-1.0,

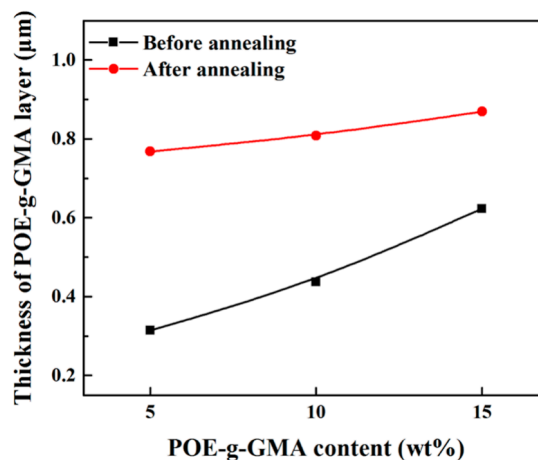


Figure 7. Thickness of the POE-g-GMA layer in the nanocomposites with 1 phr CNTs before and after annealing.

the thickness of the POE-g-GMA layer increases from 0.314 μm to 0.768 μm after annealing. As the POE-g-GMA content increases, the phase coarsening phenomenon becomes weaker. The thickness of the POE-g-GMA layer in N10-1.0 and N15-1.0, respectively, increases from 0.437 μm and 0.623 μm before annealing to 0.808 μm and 0.869 μm after annealing. The inhibition of phase coarsening can be attributed to the compatibility of POE-g-GMA to PBT/PP. The intertwining of POE-g-GMA chains located at the PBT/PP interface can effectively enhance the interfacial binding force at the interface, thereby increasing structural stability. Since the continuity in composites may be influenced by phase coarsening, the continuity of POE-g-GMA and PBT phases in the annealed composites was also tested and is reported in Table 2.

Table 2. Continuity of POE-g-GMA and PBT in the Nanocomposites with 1 phr CNTs after Annealing

sample	continuity of POE-g-GMA (%)	continuity of PBT (%)
N5-1.0	85.3	89.6
N10-1.0	89.5	91.6
N15-1.0	90.2	92.1

Combining Table 2 with Table 1, it can be seen that due to the phase agglomeration during annealing, the continuity of POE-g-GMA and PBT phases increased in three composites. However, for N5-1.0, the continuity of POE-g-GMA is less than that for N10-1.0 and N15-1.0, for its low POE-g-GMA content which cannot cover the PBT/PP interface completely.

Figure 8 shows the SEM images of PBT/POE-g-GMA/PP (60/30/10) nanocomposites with different CNT contents after annealing. All these samples were etched by xylene at 75 °C to selectively remove POE-g-GMA. Compared to the phase morphology of samples before annealing (Figure 3), N10-0 (Figure 8a) and N10-0.5 (Figure 8b) show a serious phase coarsening, while N10-1.0 (Figure 8c) and N10-2.0 (Figure 8d) show a relatively weak phase coarsening. This indicates that due to the function of CNTs as physical barriers, high

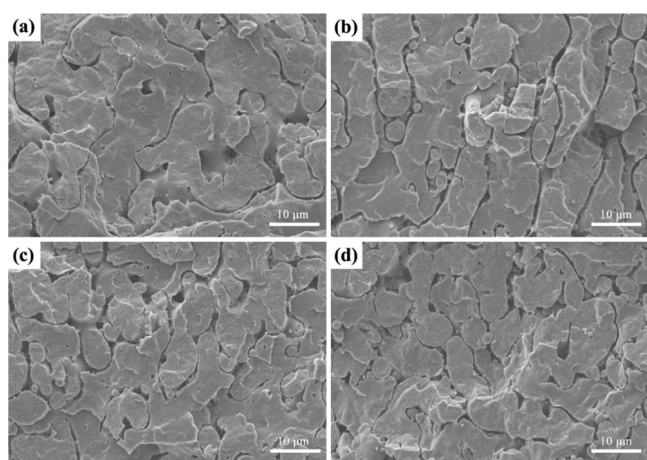


Figure 8. SEM images of the annealed N10 nanocomposites with different CNT contents: (a) N10-0; (b) N10-0.5; (c) N10-1.5; (d) N10-2.0.

CNT contents can inhibit phase coarsening during annealing. In addition, high CNT content can increase the viscosity of the POE-g-GMA phase and block polymer agglomeration.

Figure 9 displays the influence of annealing on the volume resistivity of tri-continuous nanocomposites. As seen, the

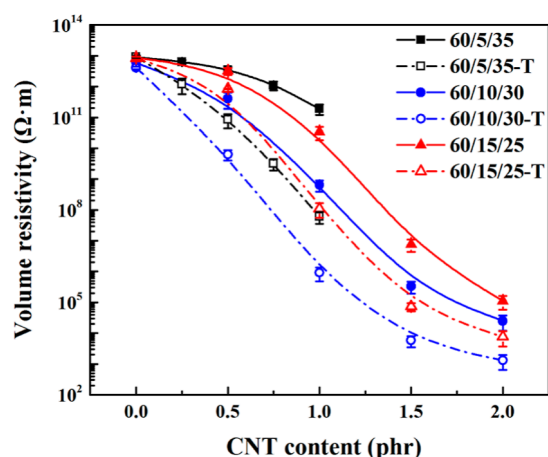


Figure 9. Volume resistivity of the nanocomposites as a function of the CNT content before and after annealing.

volume resistivity of all nanocomposites decreases after annealing, and composites of the N5 series show a sharp decline in volume resistivity. For example, the volume resistivity of N5-1.0 decreases by 4 orders of magnitude, from 1.9×10^{11} to $6.1 \times 10^7 \Omega \cdot \text{m}$. For the N10 and N15 series, the volume resistivity decreases greatly when the CNT content is below 1.0 phr while slightly with a high CNT content during annealing. The volume resistivity of N10-1.0 decreases by approximately 3 orders of magnitude, from 6.5×10^8 to $9.2 \times 10^5 \Omega \cdot \text{m}$, but that of N10-2.0 decreases by only 1 order of magnitude, from 2.5×10^4 to $1300 \Omega \cdot \text{m}$.

The improved conductivity of nanocomposites after annealing can be attributed to the reconstructed CNT networks and changes in the phase morphology (Figure 5). First, the continuity of the POE-g-GMA phase improves after annealing, facilitating a more complete CNT conductive path along the POE-g-GMA phase. Second, phase coarsening occurs by annealing attributes to decreased PBT/PP interface area

and fewer fillers needed to form a conductive path. However, decreased PBT/PP interface area after annealing results in a thicker POE-g-GMA layer, which decreases the effective concentration of CNTs in the POE-g-GMA phase and induces enhanced volume resistivity. These factors cooperatively influence the final conductivity of composites together.

For the N5 series, the continuity of POE-g-GMA improves after annealing and the POE-g-GMA layer is thin, leading to the formed CNT conductive path and enhanced conductive properties. However, due to the low POE-g-GMA content in the N5 series, the continuity of POE-g-GMA in the N5 series after annealing is still less than that in the N10 series. Therefore, the conductivity of the N5 series is lower than that of the N10 series after annealing. The N15 series after annealing have the perfect phase morphology of all series, indicating the largest PBT/PP interface area. In addition, the N15 series have the thickest POE-g-GMA layer, which means more CNTs are needed to form a well-distributed conductive path than N10, so the N15 series possess a high volume resistivity. With higher CNT contents, comprehensive CNT networks have been established in the N10 and N15 series, so annealing has little impact on the electrical conductivity, and the volume resistivity of N10 and N15 slightly reduces.

4. CONCLUSIONS

A series of PBT/POE-g-GMA/PP/CNT nanocomposites with tri-continuous structure and CNT selective distribution were obtained by melt blending in situ. The influence of CNT and POE-g-GMA content on the phase morphology and the conductive properties of nanocomposites were comprehensively studied. The stability of CNTs distributed at the PBT/PP phase interface was confirmed by thermal annealing, and the influence of thermal annealing on the microstructure morphology and conductive properties of nanocomposites was further analyzed.

In tri-continuous nanocomposites, a continuous POE-g-GMA thin layer is located at the PBT/PP interface, with CNTs selectively distributed in it. With the increase of POE-g-GMA content, the phase morphology of composites gets refined, and the thickness and continuity of the POE-g-GMA layer increase. Likewise, due to the strong physical barrier and shearing effects during melt blending, the phase morphology of composites gets refined with increasing CNT content.

Compared to PBT/PP/CNT composites, the tri-continuous composites have excellent conductive properties at a lower CNT content. The N10 series feature a better continuous POE-g-GMA layer of middle thickness, resulting in the construction of well-connected CNT conductive networks, which have a conductive percolation threshold low to 1.0 phr.

After annealing, composites remain in their tri-continuous structure with CNTs selectively distributed in a continuous POE-g-GMA layer, and the conductive networks get refined during annealing. Annealing causes phase coarsening, but POE-g-GMA and CNTs can effectively inhibit phase coarsening and stabilize phase morphology.

The evolution of the microstructure morphology during annealing has complex implications for the electrical conductivity of composites. The refined CNT networks and improved continuity of POE-g-GMA during annealing are beneficial to increase the conductive properties of composites. Phase coarsening caused by annealing could reduce the PBT/PP interface area and conductive fillers needed while thickening the POE-g-GMA layer and reducing the effective

CNT concentration. The aforementioned factors combine to reduce the volume resistivity of composites.

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Notes

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